

4.8 EXAMPLE 6

MIKROELEKTRONIKA

Module CCP1 as PWM signal generator

This example illustrates the use of CCP1 module in PWM mode. To make things more interesting, the duration of the P1A output pulses (PORTC,2) may be changed using pushbuttons symbolically marked as 'DARK' and 'BRIGHT', while the set duration is seen as binary combination on port B. The operation of this module is under control of the functions belonging to the specialized *PWM Library*. Three of them are used here:

1. **PWM1_init** has the prototype: `void Pwm1_Init(long freq);`

Parameter `freq` sets the frequency of PWM signal expressed in herz. In this example it amounts to 5kHz.

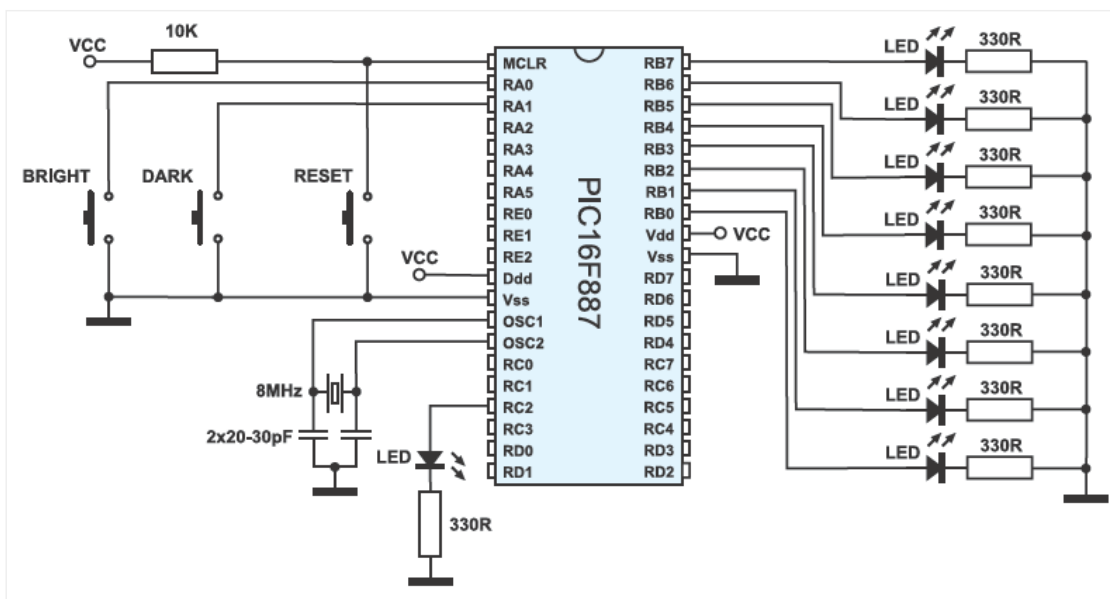
2. **PWM1 Start** has the prototype: `void Pwm1 Start(void);`

3. **PWM1_Set_Duty** has the prototype: `void Pwm1_Set_Duty(unsigned short duty_ratio);`

Parameter duty_ratio sets pulse duration in pulse sequence.

The PWM library also contains the **PWM_Stop** function used to disable this mode. Its prototype is: `void`

```
Pwm1_Stop(void);
```



```
1  /*Header*****//
2
3  unsigned short current_duty, old_duty; // Define variables
4  // current_duty and old_duty
5
6  void initMain() {
7      ANSEL = 0; // All I/O pins are configured as digital
8      ANSELH = 0;
9      PORTA = 255; // Port A initial state
10     TRISA = 255; // All port A pins are configured as inputs
11     PORTB = 0; // Initial state of port B
12     TRISB = 0; // All port B pins are configured as outputs
13     PORTC = 0; // Port C initial state
14     TRISC = 0; // All port C pins are configured as outputs
```

```
15 PWM1_Init(5000); // PWM module initialization (5KHz)
16 }
17
18 void main() {
19     initMain();
20     current_duty = 16; // Initial value of variable current_duty
21     old_duty = 0; // Reset variable old_duty
22     PWM1_Start(); // Start PWM1 module
23
24     while (1) { // Endless loop
25         if (Button(&PORTA, 0,1,1)) // If the button connected to RA0 is pressed
26             current_duty++; // increment variable current_duty
27
28         if (Button(&PORTA, 1,1,1)) // If the pressed button is connected to RA1
29             current_duty--; // decrement value current_duty
30
31         if (old_duty != current_duty) { // If current_duty and old_duty are not
32             PWM1_Set_Duty(current_duty); // equal set PWM to a new value,
33             old_duty = current_duty; // save the new value
34             PORTB = old_duty; // and show it on port B
35         }
36
37         Delay_ms(200); // 200ms delay
38     }
39 }
```

In order to make this example work properly, it is necessary to tick off the following libraries in the *Library Manager* prior to compiling:

- PWM
- Button



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